ClinSDT: LLM-Encoded Clinical Semantic Guidance for Sepsis Treatment via Decision Transformer

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Abstract

Offline sepsis treatment aims to learn treatment policies from retrospective ICU trajectories. Decision Transformer-based methods provide a natural sequence modeling framework for this task, but they mainly rely on raw physiological states and historical clinician actions. Such raw trajectory tokens may not explicitly capture clinical semantics that clinicians use in practice, such as patient severity, recent treatment response, and current risk priorities. To address this limitation, we propose **ClinSDT**, an LLM-assisted Decision Transformer framework for offline sepsis treatment. ClinSDT uses a frozen LLM as a semantic bridge rather than a treatment decision maker. It constructs three clinical semantic views from ICU trajectories, namely Profile, Context, and Advisory, and inserts their encoded representations as independent tokens into a Decision Transformer policy. To support policy refinement and rollout, ClinSDT further trains a Treatment-Response Forecaster to estimate the next physiological state, SAPS-II change, and predictive uncertainty from state–action inputs. In the second training stage, the forecaster screens local counterfactual treatment candidates, and only low-uncertainty candidates with lower predicted SAPS-II increase are distilled into the policy after semantic regeneration. During inference, ClinSDT performs closed-loop treatment rollout by refreshing semantic inputs from the currently available patient state. Experiments on MIMIC-III and MIMIC-IV show that ClinSDT achieves consistent relative improvements over representative imitation learning, offline reinforcement learning, and Decision Transformer baselines under the same rollout protocol. Code and data are available at ClinSDT.

Keywords

Sepsis Treatment, Decision Transformer, Large Language Models, Clinical Semantics, Treatment-Response Forecasting

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1 Introduction

Sepsis treatment is a critical sequential decision-making problem in intensive care medicine [4]. Patient conditions evolve over time under disease progression and treatment interventions, requiring clinicians to adjust fluid administration, vasopressor use, and other treatments based on vital signs, laboratory tests, organ function, and prior treatment history. Because online trial-and-error in real ICUs is costly and risky, retrospective ICU trajectories provide an important basis for learning clinical decision support policies [8]. Existing approaches, including behavior cloning, offline reinforcement learning, and Decision Transformer-based methods, have been used to learn treatment policies from historical trajectories [2, 11, 13]. In particular, Decision Transformer formulates offline decision-making as conditional sequence modeling and provides a natural framework for trajectory-based clinical policy learning.

Despite their usefulness, most existing methods mainly learn from raw physiological states and historical clinician actions [9]. As illustrated in Figure 4, raw ICU measurements contain observable variables, but the clinical semantics behind treatment decisions are often implicit. Clinicians usually interpret these measurements in terms of patient severity, recent treatment response, and future risk priorities. A policy trained only on raw trajectory tokens may capture temporal patterns and behavior regularities, but it may still operate under weak semantic context. This limitation is especially relevant in sepsis treatment, where similar numerical states may imply different treatment priorities depending on recent responses and risk conditions.

Large language models offer a possible way to bridge raw ICU measurements and clinical semantics. Based on this motivation, we propose **Clinical Semantic Decision Transformer (ClinSDT)**, an LLM-assisted Decision Transformer framework for offline sepsis treatment. ClinSDT uses a frozen LLM as a semantic bridge rather than a treatment decision maker. It constructs three clinical semantic views from ICU trajectories: Profile for overall patient condition, Context for recent treatment-response transitions, and

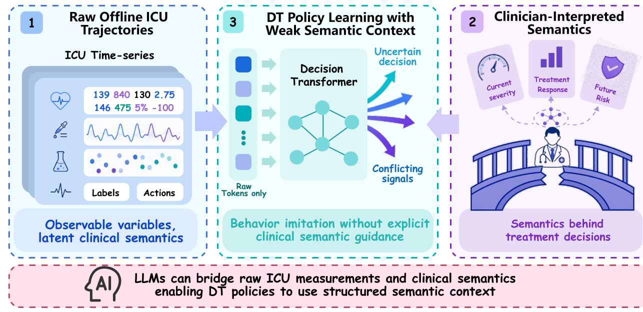

Figure 1: Motivation of ClinSDT. Raw ICU trajectories leave clinical decision semantics implicit in numerical states and actions. ClinSDT uses a frozen LLM to encode structured clinical context, while a Decision Transformer remains responsible for treatment action generation.

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ADVISORY for current risk priorities without recommending specific treatments. The encoded semantic views are inserted as independent tokens into a Decision Transformer, enabling the policy to use structured clinical context together with return-to-go, physiological states, and historical actions.

Specifically, the framework is organized around two offline training stages and one inference stage. First, ClinSDT learns a semantic-token policy from documented clinician actions and trains a forecaster to estimate short-term patient responses from state–action inputs. Second, the forecaster screens local counterfactual treatment candidates, and only low-uncertainty candidates with lower predicted SAPS-II increase are distilled into the policy after semantic regeneration. During inference, ClinSDT performs closed-loop treatment rollout by refreshing semantic inputs from the currently available patient state, generating an action with the policy, and forecasting the next state for subsequent decisions. Experiments on MIMIC-III and MIMIC-IV show that ClinSDT achieves consistent relative gains over representative imitation learning, offline reinforcement learning, and Decision Transformer baselines under the same rollout protocol.

We summarize our contributions as follows:

- We propose a semantic-guidance perspective for LLM-assisted offline clinical decision-making, where the LLM bridges raw ICU trajectories and clinical semantics while a constrained Decision Transformer remains responsible for treatment action generation.
- We construct PROFILE, CONTEXT, and ADVISORY semantic views and inject them as independent tokens into the Decision Transformer, providing structured clinical context beyond raw physiological states and historical actions.
- We introduce a two-stage training procedure that combines Semantic-Token DT learning with Treatment-Response Forecasting, followed by conservative counterfactual distillation using low-uncertainty and predicted-improving candidates.
- We evaluate ClinSDT on MIMIC-III and MIMIC-IV, showing consistent relative improvements over representative baselines under a closed-loop rollout protocol.

2 Related Work

2.1 Offline Clinical Decision Making

Offline clinical decision-making aims to learn treatment policies from retrospective clinical trajectories, avoiding the risks of online trial-and-error in real clinical environments. Existing methods mainly include behavior cloning [13], offline reinforcement learning [11], and sequence-modeling-based decision methods [2, 14]. Behavior cloning directly imitates clinician actions and is usually stable to train, but its performance is limited by the quality and coverage of historical behavior. Offline reinforcement learning further optimizes policies using return signals, but clinical applications often face distribution shift, uncertain counterfactual estimation, and limited safety guarantees [17]. Decision Transformer [2] reformulates offline decision-making as conditional sequence modeling over return-to-go, states, and actions, providing a useful framework for trajectory-based clinical policy learning [5].

Despite these advances, most offline clinical decision methods still learn mainly from low-level physiological states and historical actions. Such trajectories contain important clinical information, but the decision semantics behind treatment choices are often implicit. In sepsis treatment, clinicians usually consider patient severity, recent treatment response, and current risk priorities before selecting interventions. A policy trained only on numerical states and logged actions may capture temporal patterns, but it may not explicitly organize these treatment-oriented semantics. This paper focuses on how to provide structured clinical semantics to an offline treatment policy while retaining a constrained policy-learning framework.

2.2 LLM-assisted Clinical Decision Support

LLMs for clinical representation. Large language models have shown strong potential in medical text understanding and representation learning from electronic health records [1, 15, 16]. Recent studies have explored using LLMs to summarize patient histories, extract high-level clinical cues, and provide contextual representations for downstream clinical models. These capabilities are relevant to offline treatment learning because raw physiological trajectories often do not directly expose the clinical semantics used in decision making.

Decision agents versus semantic encoders. One line of work studies LLMs as decision-making agents that directly recommend medical interventions. However, in high-stakes settings such as sepsis management, direct treatment generation raises concerns about hallucination, controllability, action-space constraints, and offline evaluation boundaries. A more conservative alternative is to use LLMs as semantic encoders rather than treatment policies. Under this design, the LLM provides structured clinical context, while treatment actions are still generated by a downstream policy model with an explicit action space.

Semantic integration for offline policy learning. Language-derived representations can be incorporated into offline policy learning in different ways. A common approach is feature-level fusion, where semantic embeddings are concatenated with numerical states or used as auxiliary inputs. Although simple, this strategy may collapse heterogeneous clinical roles into a single representation and may not fully match the sequential structure of trajectory modeling. Decision Transformers offer a natural interface for token-level semantic integration, since clinical context, return-to-go, states, and actions can be organized as a sequence. However, how to inject heterogeneous clinical semantics into DT-style policies remains under-explored, especially when the semantic views correspond to different decision roles such as global patient status, recent treatment response, and immediate risk priorities.

Another related issue is policy refinement under offline constraints. Clinical policies trained from retrospective data may overfit observed clinician behavior, while unconstrained counterfactual updates can move the policy outside reliable data support. Prior work has studied conservative policy updates and counterfactual evaluation for offline clinical decision-making [17, 19]. In this paper, we use counterfactual signals in a restricted form: candidate actions are local, filtered by predictive uncertainty, and distilled into the policy only when they are predicted to improve SAPS-II under the learned forecaster.

Positioning of our work. ClinSDT follows the conservative semantic-encoder view of LLM-assisted clinical decision support. The LLM is not used to generate treatment actions; instead, it encodes three structured clinical views, Profile, Context, and Advisory. These views are inserted into a Decision Transformer as independent semantic tokens rather than fused into a single feature vector. Together with confidence-gated counterfactual distillation, this design provides structured clinical context for offline policy learning while preserving the constrained nature required in clinical decision settings.

3 Method

The overall framework of ClinSDT is shown in Figure 5. ClinSDT constructs Profile, Context, and Advisory semantics from offline ICU trajectories and encodes them with a frozen LLM. Training Stage I learns a semantic-token Decision Transformer policy and a Treatment-Response Forecaster for short-term patient response prediction. Training Stage II uses the forecaster to select low-uncertainty counterfactual candidates with predicted SAPS-II improvement and distills accepted candidates into the policy under regenerated semantic context. At inference time, ClinSDT conducts closed-loop treatment rollout by refreshing semantic inputs from the currently available patient state.

3.1 Problem Setup

We formulate offline sepsis treatment as a finite-horizon sequential decision-making problem. Each ICU trajectory is represented as

$$\tau = \{(s_t, a_t, r_t)\}_{t=0}^{T-1}, \tag{1}$$

where $s_t \in \mathbb{R}^{d_s}$ denotes the physiological state, $a_t \in \mathcal{A}$ is a discrete treatment action, and r_t is the clinical reward. The return-to-go is computed as

$$R_t = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}, \qquad \gamma = 0.99. \tag{2}$$

We instantiate the reward as the negative one-step SAPS-II change,

$$r_t = -\Delta_t^{\text{saps}} = -\big(\text{SAPS-II}_{t+1} - \text{SAPS-II}_t\big), \tag{3}$$

so that a decrease in SAPS-II yields positive reward and the return-to-go R_t in Eq. (41) measures the discounted cumulative SAPS-II improvement. Given the offline dataset, the goal is to learn a policy $\pi_\theta(a_t \mid H_t)$ where H_t contains the information available before selecting a_t. During training, H_t is constructed from historical trajectory segments. During rollout, it is updated from the current predicted state and the previously executed actions.

3.2 Clinical Semantic Construction

For each ICU time step, ClinSDT constructs three clinical semantic views from the available trajectory information. Profile describes the patient's overall condition, including severity, organ burden, and ventilation status. Context summarizes the recent treatment-response transition leading to the current state. Advisory describes current risk priorities, such as shock-related or organ-protection concerns, without prescribing treatment actions.

We implement these views with three rule-based constructors: a Profile constructor g_p, a transition-level Context constructor g_c, and an Advisory constructor g_a. Let c_t denote the recorded acuity record (the SAPS-II total and its organ sub-scores). For factual policy inputs, the transition available before predicting a_t is the observed incoming transition (s_{t-1}, a_{t-1}, s_t). For $t > 0$, we construct

$$x_t^p = g_p(s_t, c_t, R_t), \tag{4}$$
$$x_t^c = g_c(s_{t-1}, a_{t-1}, s_t), \tag{5}$$
$$x_t^a = g_a(s_t, c_t). \tag{6}$$

so the Profile additionally conditions on the acuity record and the return target, and the Advisory on the acuity record. Thus, the Context view uses the transition that has already occurred, rather than the current action a_t to be predicted. For $t = 0$, where no incoming transition is available, we use an initial-context template based only on the current state. At model-predicted states—counterfactual regeneration (Sec. 7.4) and rollout beyond the first step—no acuity label is available, so g_p, g_a fall back to a leakage-free state-only variant that estimates severity and organ burden from s_t alone and drops the return-target text. g_c is state-based throughout. Each constructor maps thresholded physiological features to a short qualitative description and samples among several surface realizations per clinical category; the three views are thus deterministic (up to template variants) functions of the state and expose no raw numerical values.

A frozen LLM encoder maps the constructed descriptions into semantic embeddings:

$$z_t^p = E_{\text{LLM}}(x_t^p), \tag{7}$$
$$z_t^c = E_{\text{LLM}}(x_t^c), \tag{8}$$
$$z_t^a = E_{\text{LLM}}(x_t^a). \tag{9}$$

The LLM is not used to generate treatment actions. It only provides semantic embeddings that are passed to the downstream policy model. Concretely, E_{LLM} is a frozen Qwen2.5-0.5B-Instruct model; for each view we mean-pool its last-layer hidden states over tokens to obtain an 896-dimensional embedding. The LLM is never fine-tuned.

Although the views are deterministic functions of the state, encoding them through a frozen clinical-language model maps heterogeneous descriptions into a shared, human-readable semantic space that carries external medical-language priors.

3.3 Semantic-Token DT Learning and Treatment-Response Forecasting

The first training stage contains two components. The first component trains a Semantic-Token Decision Transformer policy from documented clinical trajectories. The second component trains a Treatment-Response Forecaster that predicts short-term patient responses under state–action inputs. These two components serve different purposes: the policy models treatment decisions, while the forecaster is later used for counterfactual candidate screening and rollout.

Semantic-token policy learning. ClinSDT injects the three clinical semantic views into the Decision Transformer as independent

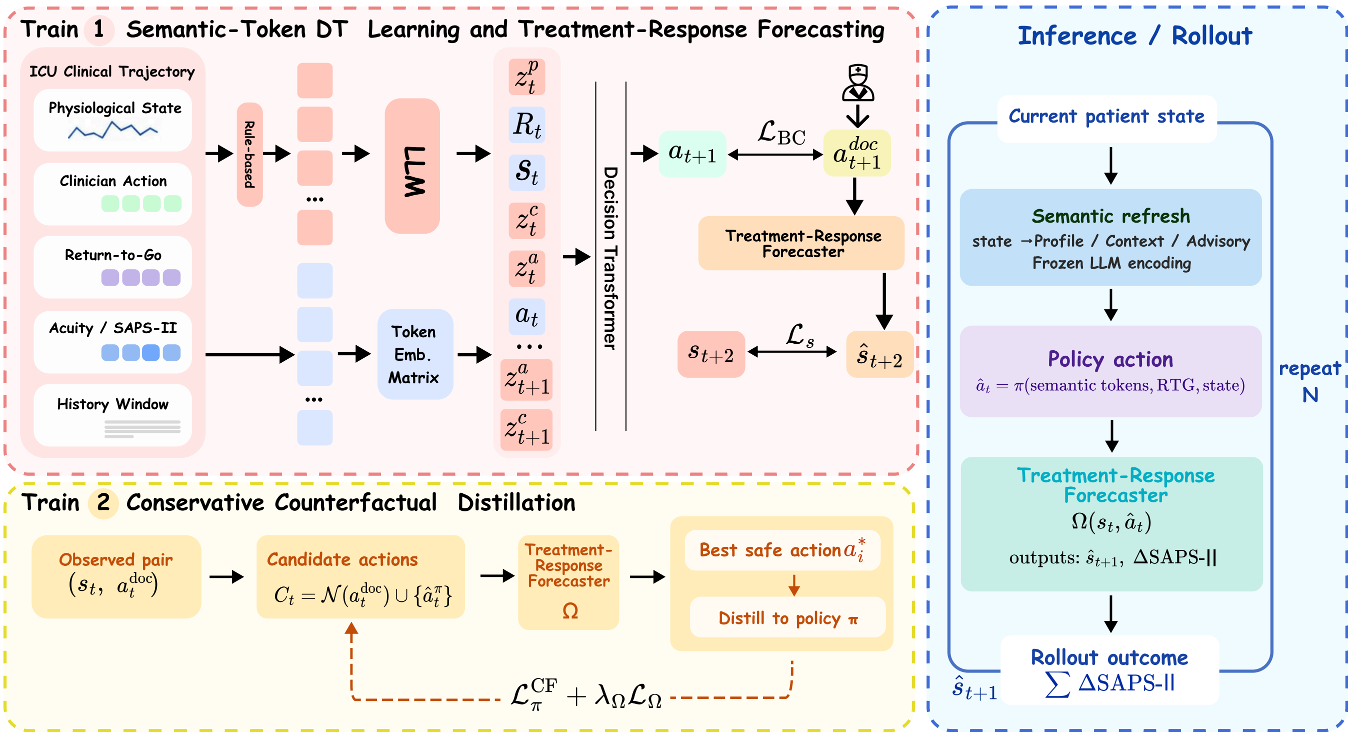

Figure 2: Overview of ClinSDT. ClinSDT encodes Profile, Context, and Advisory semantics from ICU trajectories with a frozen LLM and injects them as independent tokens into a Decision Transformer policy. Training consists of semantic-token policy learning, Treatment-Response Forecasting, and conservative counterfactual distillation with regenerated semantic inputs. During inference, the policy performs closed-loop treatment rollout by refreshing semantic inputs from the current patient state. The LLM provides semantic representations only; treatment actions are generated by the policy model.

tokens. For each time step t, we form the decision block

$$b_t = [z_t^p, R_t, z_t^c, z_t^a, s_t, a_t], \tag{10}$$

where z_t^p, z_t^c, and z_t^a denote the Profile, Context, and Advisory embeddings, respectively. The Profile token provides a global condition anchor, the return-to-go token specifies the target return condition, and the Context and Advisory tokens provide recent transition and risk-priority information before the physiological state token. Keeping these semantic views as separate tokens allows the Transformer to model their interactions with trajectory tokens without collapsing them into a single feature vector.

Given a context window of length K, the input sequence is

$$B_t = [b_{t-K+1}, \dots, b_t], \tag{11}$$

where unavailable history at the beginning of a trajectory is handled by padding and attention masking. All tokens are projected into a shared hidden space and processed by a causal Transformer. The policy head reads the hidden state at the current state token and outputs action logits:

$$\ell_t = f_\theta(h_t^s) \in \mathbb{R}^{|\mathcal{A}|}. \tag{12}$$

The action probability is given by

$$p_\theta(a \mid B_t) = \mathrm{softmax}(\ell_t)[a]. \tag{13}$$

During training, the action token is shifted in the standard autoregressive manner, so the ground-truth action a_t is not visible when predicting itself. The policy is trained by behavior cloning on documented clinician actions:

$$\mathcal{L}_{\mathrm{BC}} = -\log p_\theta(a_t^{\mathrm{doc}} \mid B_t). \tag{14}$$

Treatment-response forecasting. We train a Treatment-Response Forecaster F_ϕ, a dropout MLP, on factual transitions. A single stochastic forward pass models the next physiological state as a diagonal Gaussian and predicts the SAPS-II change:

$$F_\phi(s_t, a_t) = \left(\mu_\phi(s_t, a_t),\ \log \sigma_\phi(s_t, a_t),\ \hat{d}_t^{\mathrm{saps}}\right), \tag{15}$$

where μ_ϕ is the predicted next-state mean and $\log \sigma_\phi$ its (clamped) log standard deviation, with implied variance $\sigma_\phi^2 = \exp(2 \log \sigma_\phi)$. The forecaster takes only numerical state–action inputs and does not use LLM semantic embeddings. It is trained with factual supervision:

$$\mathcal{L}_{\mathrm{fr}}^{\mathrm{fact}} = \mathrm{NLL}\left(s_{t+1} \mid \mu_\phi, \sigma_\phi^2\right)\big|_{a_t^{\mathrm{doc}}} + \left(\hat{d}_\phi^{\mathrm{saps}}(s_t, a_t^{\mathrm{doc}}) - \Delta_t^{\mathrm{saps}}\right)^2, \tag{16}$$

where the first term is the Gaussian negative log-likelihood under the predicted mean and variance.

During counterfactual screening, we use MC dropout to estimate the reliability of each candidate transition. For each candidate action, the treatment-response forecaster is evaluated with $M = 10$ stochastic forward passes. The m-th pass produces a next-state mean $\mu_\phi^{(m)}$ and an aleatoric standard deviation $\sigma_\phi^{(m)}$. We take the averaged mean as the predicted next state:

$$\hat{s}_{t+1} = \bar{F}_\phi^s(s_t, a_t) = \frac{1}{M}\sum_{m=1}^{M} \mu_\phi^{(m)}. \tag{17}$$

To reject unreliable candidates, we further compute a scalar uncertainty score:

$$\hat{\sigma}_\phi(s_t, a_t) = \mathrm{mean}_j\left(\underbrace{\mathrm{std}_m\left[\mu_{\phi,j}^{(m)}\right]}_{\text{epistemic}} + \underbrace{\frac{1}{M}\sum_{m=1}^{M}\sigma_{\phi,j}^{(m)}}_{\text{aleatoric}}\right), \tag{18}$$

where j indexes the state dimensions. The first term captures variation across stochastic forward passes, while the second term uses the predicted aleatoric uncertainty. A counterfactual candidate is discarded if $\hat{\sigma}_\phi(s_t, a_t) \geq \sigma_{\text{thr}}$. Unless otherwise specified, F_ϕ^s denotes the MC-averaged predictor $\bar{F}_\phi^s$ in the counterfactual screening stage; the single-pass predictor is used only for optimizing the forecaster in Eq. (69).

3.4 Conservative Counterfactual Distillation

After Stage I, ClinSDT refines the policy with conservative counterfactual distillation. The procedure searches only within a local candidate set around the documented action, evaluates candidates with the Treatment-Response Forecaster, and distills an accepted candidate into the policy only when it is both low-uncertainty and predicted to yield a lower SAPS-II change (a larger short-term improvement).

Candidate construction and screening. For each factual state, we construct a candidate set using local neighbors of the documented clinician action and the current policy action:

$$\mathcal{C}_t = \mathcal{N}(a_t^{\text{doc}}) \cup \{\hat{a}_t^\pi\}, \qquad \hat{a}_t^\pi = \arg\max_a \ell_t[a], \tag{19}$$

Here $\mathcal{N}(a^{\text{doc}})$ is the local index neighborhood

$$\mathcal{N}(a^{\text{doc}}) = \left\{a^{\text{doc}}+\delta \;:\; \delta \in \{-2,\dots,2\},\ 0 \leq a^{\text{doc}}+\delta < |\mathcal{A}|\right\}, \tag{20}$$

defined on the flattened index of the 5×5 (fluid × vasopressor) action grid. Each candidate is evaluated by the forecaster. We first remove candidates with predictive uncertainty above a threshold and then select the remaining action with the lowest predicted SAPS-II change:

$$\tilde{a}_t = \arg\min_{a \in \mathcal{C}_t,\ \hat{\sigma}_\phi(s_t,a) < \sigma_{\text{thr}}} \hat{d}_\phi^{\text{saps}}(s_t, a). \tag{21}$$

This selection uses $\sigma_{\text{thr}} = 2.0$. The candidate is accepted only when it improves over the documented action under the forecaster. We additionally exclude the initial step $t = 0$, whose factual context is built from an initial-state template rather than an observed incoming transition, restricting counterfactual distillation to $t > 0$:

$$a_t^* = \begin{cases} \tilde{a}_t, & \hat{d}_\phi^{\text{saps}}(s_t, a_t^{\text{doc}}) > \hat{d}_\phi^{\text{saps}}(s_t, \tilde{a}_t) \wedge t > 0, \\ \varnothing, & \text{otherwise.} \end{cases} \tag{22}$$

Because F_ϕ predicts only one step ahead, the criterion in Eq. (60) is a greedy single-step improvement rather than a long-horizon return search; this keeps candidate evaluation local and low-variance, at the cost of not directly optimizing multi-step outcomes.

Counterfactual semantic regeneration. When a_t^* is accepted, the forecaster predicts the next state under the counterfactual action:

$$\hat{s}_{t+1}^{\text{cf}} = F_\phi^s(s_t, a_t^*). \tag{23}$$

We then regenerate the semantic inputs from the predicted counterfactual transition. The Context view is constructed from $(s_t, a_t^*, \hat{s}_{t+1}^{\text{cf}})$, and the Advisory view is constructed from the predicted next state:

$$z_t^{c,\text{cf}} = E_{\text{LLM}}\left(g_c(s_t, a_t^*, \hat{s}_{t+1}^{\text{cf}})\right), \tag{24}$$

$$z_t^{a,\text{cf}} = E_{\text{LLM}}\left(g_a(\hat{s}_{t+1}^{\text{cf}})\right). \tag{25}$$

The Profile token remains tied to the current factual state:

$$z_t^p = E_{\text{LLM}}(g_p(s_t, c_t, R_t)). \tag{26}$$

This gives the counterfactual policy input

$$B_t^{\text{cf}} = [z_t^p, R_t, z_t^{c,\text{cf}}, z_t^{a,\text{cf}}, s_t]. \tag{27}$$

The current action token is not included when predicting the counterfactual target. Using $\hat{s}_{t+1}^{\text{cf}}$ for semantic regeneration also avoids assigning the factual next state s_{t+1}, which was observed under a_t^{doc}, to the unobserved counterfactual action.

Counterfactual policy distillation. The policy is updated toward the accepted counterfactual action using the regenerated semantic input:

$$\ell_t^{\text{cf}} = f_\theta(B_t^{\text{cf}}) \in \mathbb{R}^{|\mathcal{A}|}, \tag{28}$$

$$\mathcal{L}_{\text{cf}} = \mathbb{I}[a_t^* \neq \varnothing]\,\mathrm{MSE}\left(\ell_t^{\text{cf}}, \mathbf{e}(a_t^*)\right), \tag{29}$$

where $\mathbf{e}(a_t^*)$ is the one-hot encoding of the accepted action. The loss is applied only when an accepted candidate exists. We distill toward the accepted action with an MSE-to-one-hot target rather than cross-entropy: regressing the logits to $\mathbf{e}(a_t^*)$ gives a bounded, low-variance teaching signal that we found more stable under the sparse accepted-candidate supervision.

Forecaster anchoring and bootstrap regularization. During Stage II, the forecaster continues to receive factual supervision through $\mathcal{L}_{\text{fr}}^{\text{fact}}$. For accepted candidates, we further add a confidence-gated bootstrap regularization term:

$$\mathcal{L}_{\text{boot}} = \left\|F_\phi^s(s_t, a_t^*) - \mathrm{sg}\left[\bar{F}_\phi^s(s_t, a_t^*)\right]\right\|_2^2, \tag{30}$$

where $\bar{F}_\phi^s$ is the MC-dropout averaged state prediction obtained during candidate evaluation, and $\mathrm{sg}[\cdot]$ stops gradients. Intuitively, $\mathcal{L}_{\text{boot}}$ is a confidence-gated variance-reduction anchor: since the counterfactual action has no factual next-state label, it pulls a single stochastic forward pass toward its own MC-dropout consensus $\bar{F}_\phi^s$, stabilizing the forecaster on counterfactual inputs only when uncertainty is low. The confidence score and bootstrap weight are defined as

$$c_t = \mathrm{clip}\left(1 - \frac{\hat{\sigma}_\phi(s_t, a_t^*)}{\sigma_{\text{thr}}}, 0, 1\right), \tag{31}$$

$$\alpha_t = \mathbb{I}[c_t \geq \kappa] \cdot c_t \cdot \max\left(\alpha_{\min}, 1 - \frac{n}{N}\right), \quad (32)$$

where $\kappa = 0.5$, $\alpha_{\min} = 0.3$, $N = 5 \times 10^4$, and n is the global training step.

The Stage II objective is

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{cf}} + 0.5\left(\mathcal{L}_{\text{fr}}^{\text{fact}} + \alpha_t \mathcal{L}_{\text{boot}}\right). \quad (33)$$

Thus, the policy is updated only through accepted counterfactual candidates, while the forecaster remains anchored to factual transitions and receives bootstrap regularization only under sufficient confidence.

3.5 Inference Closed-Loop Semantic-Guided Treatment Rollout

At inference time, ClinSDT generates treatment actions in a closed-loop rollout. The current state is first converted into Profile, Context, and Advisory descriptions according to the information available at that step. These descriptions are encoded by the frozen LLM and passed to the policy model. The current action token is absent, and the policy selects the treatment action by

$$\hat{a}_t = \arg\max_{a \in \mathcal{A}} \ell_t[a], \quad (34)$$

from the input

$$[z_t^p, R_t, z_t^c, z_t^a, s_t]. \quad (35)$$

For rollout beyond the initially observed state, future acuity annotations are not available. Therefore, ClinSDT uses a state-only semantic template for predicted states, constructed from the physiological state variables rather than future labels. Given the current rollout state $\hat{s}_k$ and policy action $\hat{a}_k$, the forecaster updates the state as

$$\hat{s}_{k+1} = F_\phi^s(\hat{s}_k, \hat{a}_k). \quad (36)$$

The updated state is then used to refresh the semantic inputs for the next decision step. This procedure is repeated for the rollout horizon. Throughout inference, the LLM only encodes semantic descriptions, while treatment actions are generated by the policy model.

4 Experiments

We evaluate ClinSDT from four perspectives that follow the motivation of the paper: whether structured clinical semantics improve offline policy learning, which semantic designs contribute to the gain, whether semantic tokens participate in policy learning and counterfactual refinement, and how the rollout-based evidence should be interpreted.

RQ1: Does structured LLM-encoded clinical semantics improve offline sepsis treatment policies?

RQ2: Which semantic factors matter: the clinical views themselves or their token organization?

RQ3: Does Stage II provide additional counterfactual refinement, and are the semantic tokens actively used by the policy?

RQ4: What are the reliability boundaries of model-based rollout evaluation?

Datasets. We evaluate on offline ICU sepsis treatment trajectories from MIMIC-III and MIMIC-IV [6, 7]. Each trajectory contains physiological measurements, historical treatment actions, and subsequent patient-condition changes. Following common sepsis treatment settings, each state is represented by a 45-dimensional physiological feature vector, and actions are discretized according to intravenous fluid and vasopressor levels. For subgroup analysis, patients are divided into **HIGH**, **MID**, and **LOW** acuity groups according to initial SAPS-II scores. Because SAPS-II distributions differ across datasets, we use dataset-specific thresholds while keeping the same three-level acuity structure.

Evaluation metric. We use SAPS-II change as the proxy outcome for model-based rollout evaluation. Let $q_{i,t}$ denote the predicted SAPS-II score of trajectory i at rollout step t. For horizon K, the cumulative SAPS-II change is

$$\Delta\text{SAPS}_{i,K} = \sum_{t=1}^{K}(q_{i,t} - q_{i,t-1}) = q_{i,K} - q_{i,0}. \quad (37)$$

Since lower SAPS-II indicates improved patient condition, we report SAPS-II improvement:

$$\text{Imp}_{i,K} = -\Delta\text{SAPS}_{i,K}. \quad (38)$$

The final score is averaged over the test set $\mathcal{D}$:

$$\text{Imp}_K = \frac{1}{|\mathcal{D}|}\sum_{i \in \mathcal{D}} \text{Imp}_{i,K}. \quad (39)$$

Higher values indicate better predicted outcomes. We report both 5-step and 10-step rollouts. All policies are evaluated with the same learned rollout environment and the same initial test trajectories. Therefore, this metric should be interpreted as controlled offline evidence for relative policy comparison, rather than as direct evidence of clinical efficacy.

Baselines. We compare ClinSDT with representative baselines from imitation learning, value-based RL, offline RL, and sequence-modeling-based policy learning. **BC** [13] imitates logged clinician actions. **DQN** [12] is a value-based RL baseline trained under the same offline data and rollout protocol. **BCQ** [3] constrains action selection to the support of the behavior data. **IQL** [10] performs implicit offline policy optimization without explicit out-of-distribution action evaluation. **DT** [2] is a return-conditioned Decision Transformer trained on return, state, and action sequences.

4.1 Overall Effectiveness of Structured Clinical Semantics (RQ1)

RQ1 evaluates whether adding structured clinical semantics to a DT-style policy improves offline sepsis treatment decision-making. Table 2 reports SAPS-II improvement on MIMIC-III and MIMIC-IV under 5-step and 10-step rollouts.

ClinSDT achieves the best ALL-patient performance on both datasets and both rollout horizons. On MIMIC-III, the strongest baseline reaches +0.57 under the 5-step rollout and +2.94 under the 10-step rollout, while ClinSDT achieves +1.36 and +8.75, respectively. The gain is more visible in the longer rollout, where policy decisions interact with evolving patient states over multiple steps. Under the 10-step setting on MIMIC-III, ClinSDT also

Table 1: Main rollout results on MIMIC-III and MIMIC-IV. We compare ClinSDT with representative baselines from imitation learning, offline RL, value-based RL, and Decision Transformer-style policy learning. The metric is SAPS-II improvement, defined as the negative cumulative SAPS-II change over rollout trajectories. Higher values indicate better predicted outcomes. Bold indicates the best result in each column, and underlining indicates the second-best result.

Dataset	Model	5-step rollout				10-step rollout			
		HIGH	MID	LOW	ALL	HIGH	MID	LOW	ALL
MIMIC-III	BC	3.24	-1.51	-5.59	-0.59	5.39	-1.55	-7.80	-0.22
	BCQ	3.17	-1.30	-3.88	-0.35	5.99	-1.17	-5.59	0.34
	DQN	6.42	-1.05	-5.03	0.57	13.17	0.10	-6.74	2.94
	DT	**7.01**	-1.57	-5.93	0.30	12.96	-1.40	-8.98	1.72
	IQL	2.86	-2.13	-6.13	-1.14	4.39	-2.58	-9.25	-1.28
	Ours	6.91	**-0.44**	**-1.43**	**1.36**	**17.74**	**5.16**	**10.77**	**8.75**
MIMIC-IV	BC	0.55	-0.80	-0.62	-0.26	0.34	-1.61	-1.23	-0.79
	BCQ	1.82	-0.66	-0.93	0.13	**2.59**	-1.02	-1.53	0.10
	DQN	1.38	-0.43	-0.50	0.19	1.97	-0.95	-1.06	0.06
	DT	**2.17**	-0.74	-0.95	0.23	2.48	-1.90	-2.18	-0.43
	IQL	0.17	-0.84	-0.77	-0.46	-1.23	-2.15	-1.62	-1.65
	Ours	0.65	**0.21**	**0.17**	**0.35**	0.76	**0.30**	**0.27**	**0.45**

achieves the best result in all three acuity groups, suggesting that the overall gain is not driven by a single subgroup.

On MIMIC-IV, the absolute improvements are smaller, but ClinSDT still achieves the best ALL score under both horizons, reaching +0.35 and +0.45 for 5-step and 10-step rollouts. This cross-dataset trend supports the main hypothesis of the paper: raw physiological trajectories alone may leave important decision semantics implicit, while structured clinical semantic tokens provide useful context for sequential policy learning.

These results should be interpreted as relative policy-quality evidence under a shared learned evaluator. They do not establish real-world clinical effectiveness, but they indicate that the proposed semantic-guided policy consistently compares favorably with imitation learning, offline RL, and standard DT baselines in the controlled rollout setting. Figure 6 further summarizes the ALL-patient results across datasets and rollout horizons.

4.2 Semantic Factors and Token Organization (RQ2)

RQ2 examines which part of the semantic design contributes to the observed gains. We separate two factors: the content of the clinical semantic views and the way these views are organized in the policy input. This distinction is important because ClinSDT does not simply concatenate extra features; it inserts role-specific semantic views as separate tokens.

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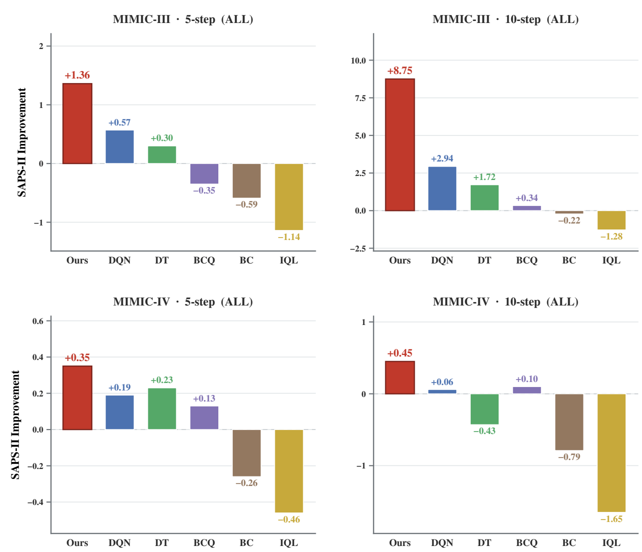

Figure 3: Main rollout results on MIMIC-III and MIMIC-IV. ClinSDT achieves the best ALL-patient performance under both 5-step and 10-step rollouts, with the clearest gain on the longer MIMIC-III rollout.

ClinSDT: LLM-Encoded Clinical Semantic Guidance for Sepsis Treatment via Decision Transformer

Anonymous Author(s)

Abstract

Offline sepsis treatment aims to learn treatment policies from retrospective ICU trajectories. Decision Transformer-based methods provide a natural sequence modeling framework for this task, but they mainly rely on raw physiological states and historical clinician actions. Such raw trajectory tokens may not explicitly capture clinical semantics that clinicians use in practice, such as patient severity, recent treatment response, and current risk priorities. To address this limitation, we propose **ClinSDT**, an LLM-assisted Decision Transformer framework for offline sepsis treatment. ClinSDT uses a frozen LLM as a semantic bridge rather than a treatment decision maker. It constructs three clinical semantic views from ICU trajectories, namely PROFILE, CONTEXT, and ADVISORY, and inserts their encoded representations as independent tokens into a Decision Transformer policy. To support policy refinement and rollout, ClinSDT further trains a Treatment-Response Forecaster to estimate the next physiological state, SAPS-II change, and predictive uncertainty from state–action inputs. In the second training stage, the forecaster screens local counterfactual treatment candidates, and only low-uncertainty candidates with lower predicted SAPS-II increase are distilled into the policy after semantic regeneration. During inference, ClinSDT performs closed-loop treatment rollout by refreshing semantic inputs from the currently available patient state. Experiments on MIMIC-III and MIMIC-IV show that ClinSDT achieves consistent relative improvements over representative imitation learning, offline reinforcement learning, and Decision Transformer baselines under the same rollout protocol. Code and data are available at ClinSDT.

Keywords

Sepsis Treatment, Decision Transformer, Large Language Models, Clinical Semantics, Treatment-Response Forecasting

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5 Introduction

Sepsis treatment is a critical sequential decision-making problem in intensive care medicine [4]. Patient conditions evolve over time

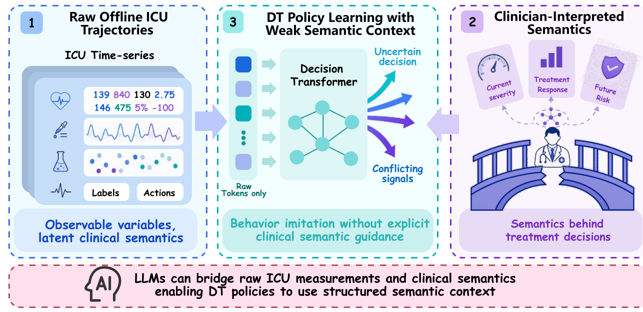

Figure 4: Motivation of ClinSDT. Raw ICU trajectories leave clinical decision semantics implicit in numerical states and actions. ClinSDT uses a frozen LLM to encode structured clinical context, while a Decision Transformer remains responsible for treatment action generation.

under disease progression and treatment interventions, requiring clinicians to adjust fluid administration, vasopressor use, and other treatments based on vital signs, laboratory tests, organ function, and prior treatment history. Because online trial-and-error in real ICUs is costly and risky, retrospective ICU trajectories provide an important basis for learning clinical decision support policies [8]. Existing approaches, including behavior cloning, offline reinforcement learning, and Decision Transformer-based methods, have been used to learn treatment policies from historical trajectories [2, 11, 13]. In particular, Decision Transformer formulates offline decision-making as conditional sequence modeling and provides a natural framework for trajectory-based clinical policy learning.

Despite their usefulness, most existing methods mainly learn from raw physiological states and historical clinician actions [9]. As illustrated in Figure 4, raw ICU measurements contain observable variables, but the clinical semantics behind treatment decisions are often implicit. Clinicians usually interpret these measurements in terms of patient severity, recent treatment response, and future risk priorities. A policy trained only on raw trajectory tokens may capture temporal patterns and behavior regularities, but it may still operate under weak semantic context. This limitation is especially relevant in sepsis treatment, where similar numerical states may imply different treatment priorities depending on recent responses and risk conditions.

Large language models offer a possible way to bridge raw ICU measurements and clinical semantics. Based on this motivation, we propose **Clinical Semantic Decision Transformer (ClinSDT)**, an LLM-assisted Decision Transformer framework for offline sepsis treatment. ClinSDT uses a frozen LLM as a semantic bridge rather than a treatment decision maker. It constructs three clinical semantic views from ICU trajectories: PROFILE for overall patient condition, CONTEXT for recent treatment-response transitions, and

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Advisory for current risk priorities without recommending specific treatments. The encoded semantic views are inserted as independent tokens into a Decision Transformer, enabling the policy to use structured clinical context together with return-to-go, physiological states, and historical actions.

Specifically, the framework is organized around two offline training stages and one inference stage. First, ClinSDT learns a semantic-token policy from documented clinician actions and trains a forecaster to estimate short-term patient responses from state–action inputs. Second, the forecaster screens local counterfactual treatment candidates, and only low-uncertainty candidates with lower predicted SAPS-II increase are distilled into the policy after semantic regeneration. During inference, ClinSDT performs closed-loop treatment rollout by refreshing semantic inputs from the currently available patient state, generating an action with the policy, and forecasting the next state for subsequent decisions. Experiments on MIMIC-III and MIMIC-IV show that ClinSDT achieves consistent relative gains over representative imitation learning, offline reinforcement learning, and Decision Transformer baselines under the same rollout protocol.

We summarize our contributions as follows:

- We propose a semantic-guidance perspective for LLM-assisted offline clinical decision-making, where the LLM bridges raw ICU trajectories and clinical semantics while a constrained Decision Transformer remains responsible for treatment action generation.
- We construct Profile, Context, and Advisory semantic views and inject them as independent tokens into the Decision Transformer, providing structured clinical context beyond raw physiological states and historical actions.
- We introduce a two-stage training procedure that combines Semantic-Token DT learning with Treatment-Response Forecasting, followed by conservative counterfactual distillation using low-uncertainty and predicted-improving candidates.
- We evaluate ClinSDT on MIMIC-III and MIMIC-IV, showing consistent relative improvements over representative baselines under a closed-loop rollout protocol.

6 Related Work

6.1 Offline Clinical Decision Making

Offline clinical decision-making aims to learn treatment policies from retrospective clinical trajectories, avoiding the risks of online trial-and-error in real clinical environments. Existing methods mainly include behavior cloning [13], offline reinforcement learning [11], and sequence-modeling-based decision methods [2, 14]. Behavior cloning directly imitates clinician actions and is usually stable to train, but its performance is limited by the quality and coverage of historical behavior. Offline reinforcement learning further optimizes policies using return signals, but clinical applications often face distribution shift, uncertain counterfactual estimation, and limited safety guarantees [17]. Decision Transformer [2] reformulates offline decision-making as conditional sequence modeling over return-to-go, states, and actions, providing a useful framework for trajectory-based clinical policy learning [5].

Despite these advances, most offline clinical decision methods still learn mainly from low-level physiological states and historical actions. Such trajectories contain important clinical information, but the decision semantics behind treatment choices are often implicit. In sepsis treatment, clinicians usually consider patient severity, recent treatment response, and current risk priorities before selecting interventions. A policy trained only on numerical states and logged actions may capture temporal patterns, but it may not explicitly organize these treatment-oriented semantics. This paper focuses on how to provide structured clinical semantics to an offline treatment policy while retaining a constrained policy-learning framework.

6.2 LLM-assisted Clinical Decision Support

LLMs for clinical representation. Large language models have shown strong potential in medical text understanding and representation learning from electronic health records [1, 15, 16]. Recent studies have explored using LLMs to summarize patient histories, extract high-level clinical cues, and provide contextual representations for downstream clinical models. These capabilities are relevant to offline treatment learning because raw physiological trajectories often do not directly expose the clinical semantics used in decision making.

Decision agents versus semantic encoders. One line of work studies LLMs as decision-making agents that directly recommend medical interventions. However, in high-stakes settings such as sepsis management, direct treatment generation raises concerns about hallucination, controllability, action-space constraints, and offline evaluation boundaries. A more conservative alternative is to use LLMs as semantic encoders rather than treatment policies. Under this design, the LLM provides structured clinical context, while treatment actions are still generated by a downstream policy model with an explicit action space.

Semantic integration for offline policy learning. Language-derived representations can be incorporated into offline policy learning in different ways. A common approach is feature-level fusion, where semantic embeddings are concatenated with numerical states or used as auxiliary inputs. Although simple, this strategy may collapse heterogeneous clinical roles into a single representation and may not fully match the sequential structure of trajectory modeling. Decision Transformers offer a natural interface for token-level semantic integration, since clinical context, return-to-go, states, and actions can be organized as a sequence. However, how to inject heterogeneous clinical semantics into DT-style policies remains under-explored, especially when the semantic views correspond to different decision roles such as global patient status, recent treatment response, and immediate risk priorities.

Another related issue is policy refinement under offline constraints. Clinical policies trained from retrospective data may overfit observed clinician behavior, while unconstrained counterfactual updates can move the policy outside reliable data support. Prior work has studied conservative policy updates and counterfactual evaluation for offline clinical decision-making [17, 19]. In this paper, we use counterfactual signals in a restricted form: candidate actions are local, filtered by predictive uncertainty, and distilled into the policy only when they are predicted to improve SAPS-II under the learned forecaster.

Positioning of our work. ClinSDT follows the conservative semantic-encoder view of LLM-assisted clinical decision support. The LLM is not used to generate treatment actions; instead, it encodes three structured clinical views, Profile, Context, and Advisory. These views are inserted into a Decision Transformer as independent semantic tokens rather than fused into a single feature vector. Together with confidence-gated counterfactual distillation, this design provides structured clinical context for offline policy learning while preserving the constrained nature required in clinical decision settings.

7 Method

The overall framework of ClinSDT is shown in Figure 5. ClinSDT constructs Profile, Context, and Advisory semantics from offline ICU trajectories and encodes them with a frozen LLM. Training Stage I learns a semantic-token Decision Transformer policy and a Treatment-Response Forecaster for short-term patient response prediction. Training Stage II uses the forecaster to select low-uncertainty counterfactual candidates with predicted SAPS-II improvement and distills accepted candidates into the policy under regenerated semantic context. At inference time, ClinSDT conducts closed-loop treatment rollout by refreshing semantic inputs from the currently available patient state.

7.1 Problem Setup

We formulate offline sepsis treatment as a finite-horizon sequential decision-making problem. Each ICU trajectory is represented as

$$\tau = \{(s_t, a_t, r_t)\}_{t=0}^{T-1}, \tag{40}$$

where $s_t \in \mathbb{R}^{d_s}$ denotes the physiological state, $a_t \in \mathcal{A}$ is a discrete treatment action, and r_t is the clinical reward. The return-to-go is computed as

$$R_t = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}, \qquad \gamma = 0.99. \tag{41}$$

We instantiate the reward as the negative one-step SAPS-II change,

$$r_t = -\Delta_t^{\text{saps}} = -\big(\text{SAPS-II}_{t+1} - \text{SAPS-II}_t\big), \tag{42}$$

so that a decrease in SAPS-II yields positive reward and the return-to-go R_t in Eq. (41) measures the discounted cumulative SAPS-II improvement. Given the offline dataset, the goal is to learn a policy $\pi_\theta(a_t \mid H_t)$ where H_t contains the information available before selecting a_t. During training, H_t is constructed from historical trajectory segments. During rollout, it is updated from the current predicted state and the previously executed actions.

7.2 Clinical Semantic Construction

For each ICU time step, ClinSDT constructs three clinical semantic views from the available trajectory information. Profile describes the patient's overall condition, including severity, organ burden, and ventilation status. Context summarizes the recent treatment-response transition leading to the current state. Advisory describes current risk priorities, such as shock-related or organ-protection concerns, without prescribing treatment actions.

We implement these views with three rule-based constructors: a Profile constructor g_p, a transition-level Context constructor g_c, and an Advisory constructor g_a. Let c_t denote the recorded acuity record (the SAPS-II total and its organ sub-scores). For factual policy inputs, the transition available before predicting a_t is the observed incoming transition (s_{t-1}, a_{t-1}, s_t). For $t > 0$, we construct

$$x_t^p = g_p(s_t, c_t, R_t), \tag{43}$$

$$x_t^c = g_c(s_{t-1}, a_{t-1}, s_t), \tag{44}$$

$$x_t^a = g_a(s_t, c_t). \tag{45}$$

so the Profile additionally conditions on the acuity record and the return target, and the Advisory on the acuity record. Thus, the Context view uses the transition that has already occurred, rather than the current action a_t to be predicted. For $t = 0$, where no incoming transition is available, we use an initial-context template based only on the current state. At model-predicted states—counterfactual regeneration (Sec. 7.4) and rollout beyond the first step—no acuity label is available, so g_p, g_a fall back to a leakage-free state-only variant that estimates severity and organ burden from s_t alone and drops the return-target text. g_c is state-based throughout. Each constructor maps thresholded physiological features to a short qualitative description and samples among several surface realizations per clinical category; the three views are thus deterministic (up to template variants) functions of the state and expose no raw numerical values.

A frozen LLM encoder maps the constructed descriptions into semantic embeddings:

$$z_t^p = E_{\text{LLM}}(x_t^p), \tag{46}$$

$$z_t^c = E_{\text{LLM}}(x_t^c), \tag{47}$$

$$z_t^a = E_{\text{LLM}}(x_t^a). \tag{48}$$

The LLM is not used to generate treatment actions. It only provides semantic embeddings that are passed to the downstream policy model. Concretely, E_{LLM} is a frozen Qwen2.5-0.5B-Instruct model; for each view we mean-pool its last-layer hidden states over tokens to obtain an 896-dimensional embedding. The LLM is never fine-tuned.

Although the views are deterministic functions of the state, encoding them through a frozen clinical-language model maps heterogeneous descriptions into a shared, human-readable semantic space that carries external medical-language priors.

7.3 Semantic-Token DT Learning and Treatment-Response Forecasting

The first training stage contains two components. The first component trains a Semantic-Token Decision Transformer policy from documented clinical trajectories. The second component trains a Treatment-Response Forecaster that predicts short-term patient responses under state–action inputs. These two components serve different purposes: the policy models treatment decisions, while the forecaster is later used for counterfactual candidate screening and rollout.

Semantic-token policy learning. ClinSDT injects the three clinical semantic views into the Decision Transformer as independent

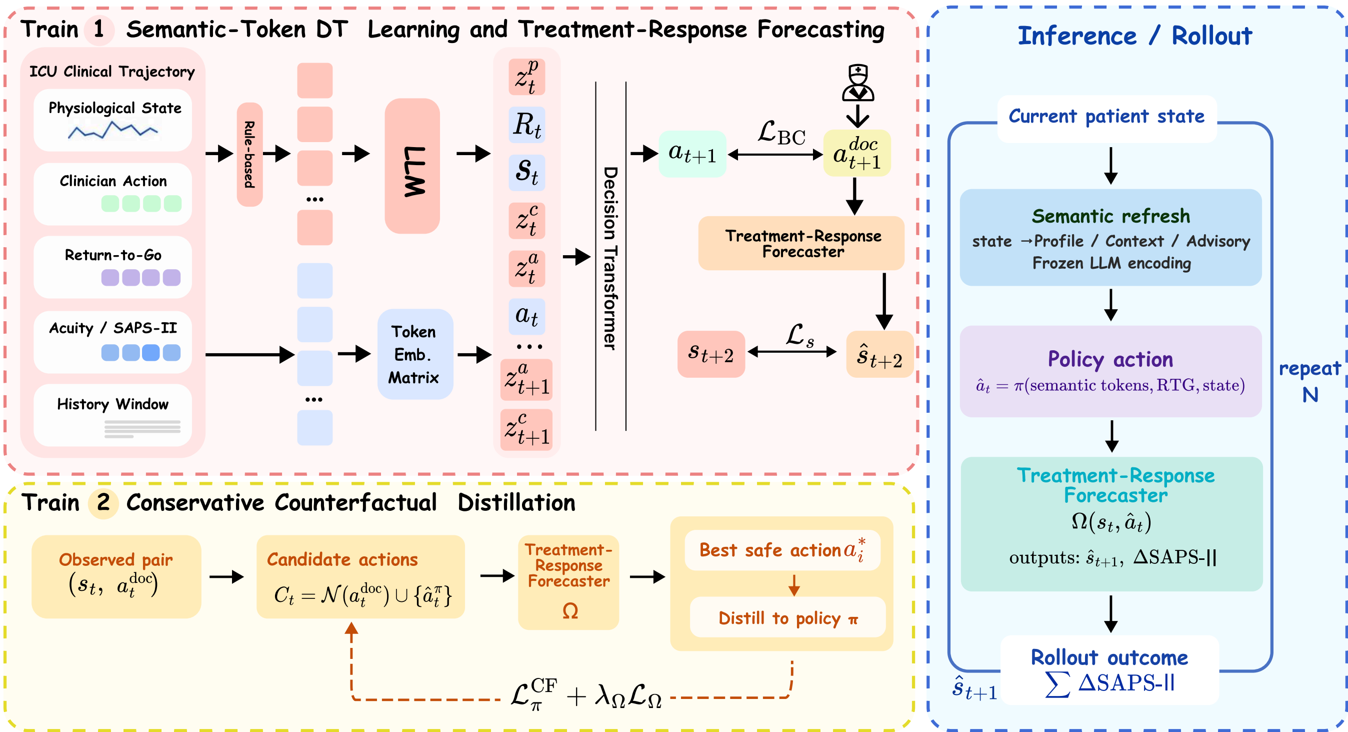

Figure 5: Overview of ClinSDT. ClinSDT encodes Profile, Context, and Advisory semantics from ICU trajectories with a frozen LLM and injects them as independent tokens into a Decision Transformer policy. Training consists of semantic-token policy learning, Treatment-Response Forecasting, and conservative counterfactual distillation with regenerated semantic inputs. During inference, the policy performs closed-loop treatment rollout by refreshing semantic inputs from the current patient state. The LLM provides semantic representations only; treatment actions are generated by the policy model.

tokens. For each time step t, we form the decision block

$$b_t = [z_t^p, R_t, z_t^c, z_t^a, s_t, a_t], \tag{49}$$

where z_t^p, z_t^c, and z_t^a denote the Profile, Context, and Advisory embeddings, respectively. The Profile token provides a global condition anchor, the return-to-go token specifies the target return condition, and the Context and Advisory tokens provide recent transition and risk-priority information before the physiological state token. Keeping these semantic views as separate tokens allows the Transformer to model their interactions with trajectory tokens without collapsing them into a single feature vector.

Given a context window of length K, the input sequence is

$$B_t = [b_{t-K+1}, \ldots, b_t], \tag{50}$$

where unavailable history at the beginning of a trajectory is handled by padding and attention masking. All tokens are projected into a shared hidden space and processed by a causal Transformer. The policy head reads the hidden state at the current state token and outputs action logits:

$$\ell_t = f_\theta(h_t^s) \in \mathbb{R}^{|\mathcal{A}|}. \tag{51}$$

The action probability is given by

$$p_\theta(a \mid B_t) = \mathrm{softmax}(\ell_t)[a]. \tag{52}$$

During training, the action token is shifted in the standard autoregressive manner, so the ground-truth action a_t is not visible when predicting itself. The policy is trained by behavior cloning on documented clinician actions:

$$\mathcal{L}_{\mathrm{BC}} = -\log p_\theta(a_t^{\mathrm{doc}} \mid B_t). \tag{53}$$

Treatment-response forecasting. We train a Treatment-Response Forecaster F_ϕ, a dropout MLP, on factual transitions. A single stochastic forward pass models the next physiological state as a diagonal Gaussian and predicts the SAPS-II change:

$$F_\phi(s_t, a_t) = \big(\mu_\phi(s_t, a_t),\ \log\sigma_\phi(s_t, a_t),\ \hat{d}_t^{\mathrm{saps}}\big), \tag{54}$$

where μ_ϕ is the predicted next-state mean and $\log\sigma_\phi$ its (clamped) log standard deviation, with implied variance $\sigma_\phi^2 = \exp(2\log\sigma_\phi)$. The forecaster takes only numerical state–action inputs and does not use LLM semantic embeddings. It is trained with factual supervision:

$$\mathcal{L}_{\mathrm{fr}}^{\mathrm{fact}} = \mathrm{NLL}\big(s_{t+1} \mid \mu_\phi, \sigma_\phi^2\big)\big|_{a_t^{\mathrm{doc}}} + \big(\hat{d}_\phi^{\mathrm{saps}}(s_t, a_t^{\mathrm{doc}}) - \Delta_t^{\mathrm{saps}}\big)^2, \tag{55}$$

where the first term is the Gaussian negative log-likelihood under the predicted mean and variance.

During counterfactual screening, we use MC dropout to estimate the reliability of each candidate transition. For each candidate action, the treatment-response forecaster is evaluated with $M = 10$ stochastic forward passes. The m-th pass produces a next-state mean $\mu_\phi^{(m)}$ and an aleatoric standard deviation $\sigma_\phi^{(m)}$. We take the averaged mean as the predicted next state:

$$\hat{s}_{t+1} = \bar{F}_\phi^s(s_t, a_t) = \frac{1}{M}\sum_{m=1}^{M} \mu_\phi^{(m)}. \tag{56}$$

To reject unreliable candidates, we further compute a scalar uncertainty score:

$$\hat{\sigma}_\phi(s_t, a_t) = \mathrm{mean}_j \left(\underbrace{\mathrm{std}_m\left[\mu_{\phi,j}^{(m)}\right]}_{\text{epistemic}} + \underbrace{\frac{1}{M}\sum_{m=1}^{M} \sigma_{\phi,j}^{(m)}}_{\text{aleatoric}} \right), \tag{57}$$

where j indexes the state dimensions. The first term captures variation across stochastic forward passes, while the second term uses the predicted aleatoric uncertainty. A counterfactual candidate is discarded if $\hat{\sigma}_\phi(s_t, a_t) \geq \sigma_{\text{thr}}$. Unless otherwise specified, F_ϕ^s denotes the MC-averaged predictor $\bar{F}_\phi^s$ in the counterfactual screening stage; the single-pass predictor is used only for optimizing the forecaster in Eq. (69).

7.4 Conservative Counterfactual Distillation

After Stage I, ClinSDT refines the policy with conservative counterfactual distillation. The procedure searches only within a local candidate set around the documented action, evaluates candidates with the Treatment-Response Forecaster, and distills an accepted candidate into the policy only when it is both low-uncertainty and predicted to yield a lower SAPS-II change (a larger short-term improvement).

Candidate construction and screening. For each factual state, we construct a candidate set using local neighbors of the documented clinician action and the current policy action:

$$\mathcal{C}_t = \mathcal{N}(a_t^{\text{doc}}) \cup \{\hat{a}_t^\pi\}, \qquad \hat{a}_t^\pi = \arg\max_a \ell_t[a], \tag{58}$$

Here $\mathcal{N}(a^{\text{doc}})$ is the local index neighborhood

$$\mathcal{N}(a^{\text{doc}}) = \left\{ a^{\text{doc}} + \delta \;:\; \delta \in \{-2, \dots, 2\},\ 0 \leq a^{\text{doc}} + \delta < |\mathcal{A}| \right\}, \tag{59}$$

defined on the flattened index of the 5×5 (fluid $\times$ vasopressor) action grid. Each candidate is evaluated by the forecaster. We first remove candidates with predictive uncertainty above a threshold and then select the remaining action with the lowest predicted SAPS-II change:

$$\tilde{a}_t = \arg \min_{a \in \mathcal{C}_t,\ \hat{\sigma}_\phi(s_t, a) < \sigma_{\text{thr}}} \hat{d}_\phi^{\text{saps}}(s_t, a). \tag{60}$$

This selection uses $\sigma_{\text{thr}} = 2.0$. The candidate is accepted only when it improves over the documented action under the forecaster. We additionally exclude the initial step $t = 0$, whose factual context is built from an initial-state template rather than an observed incoming transition, restricting counterfactual distillation to $t > 0$:

$$a_t^* = \begin{cases} \tilde{a}_t, & \hat{d}_\phi^{\text{saps}}(s_t, a_t^{\text{doc}}) > \hat{d}_\phi^{\text{saps}}(s_t, \tilde{a}_t) \ \wedge\ t > 0, \\ \varnothing, & \text{otherwise}. \end{cases} \tag{61}$$

Because F_ϕ predicts only one step ahead, the criterion in Eq. (60) is a greedy single-step improvement rather than a long-horizon return search; this keeps candidate evaluation local and low-variance, at the cost of not directly optimizing multi-step outcomes.

Counterfactual semantic regeneration. When a_t^* is accepted, the forecaster predicts the next state under the counterfactual action:

$$\hat{s}_{t+1}^{\text{cf}} = F_\phi^s(s_t, a_t^*). \tag{62}$$

We then regenerate the semantic inputs from the predicted counterfactual transition. The Context view is constructed from $(s_t, a_t^*, \hat{s}_{t+1}^{\text{cf}})$, and the Advisory view is constructed from the predicted next state:

$$z_t^{c,\text{cf}} = E_{\text{LLM}}\left(g_c(s_t, a_t^*, \hat{s}_{t+1}^{\text{cf}})\right), \tag{63}$$

$$z_t^{a,\text{cf}} = E_{\text{LLM}}\left(g_a(\hat{s}_{t+1}^{\text{cf}})\right). \tag{64}$$

The Profile token remains tied to the current factual state:

$$z_t^p = E_{\text{LLM}}(g_p(s_t, c_t, R_t)). \tag{65}$$

This gives the counterfactual policy input

$$B_t^{\text{cf}} = [z_t^p, R_t, z_t^{c,\text{cf}}, z_t^{a,\text{cf}}, s_t]. \tag{66}$$

The current action token is not included when predicting the counterfactual target. Using $\hat{s}_{t+1}^{\text{cf}}$ for semantic regeneration also avoids assigning the factual next state s_{t+1}, which was observed under a_t^{doc}, to the unobserved counterfactual action.

Counterfactual policy distillation. The policy is updated toward the accepted counterfactual action using the regenerated semantic input:

$$\ell_t^{\text{cf}} = f_\theta(B_t^{\text{cf}}) \in \mathbb{R}^{|\mathcal{A}|}, \tag{67}$$

$$\mathcal{L}_{\text{cf}} = \mathbb{I}[a_t^* \neq \varnothing]\, \mathrm{MSE}\left(\ell_t^{\text{cf}}, \mathbf{e}(a_t^*)\right), \tag{68}$$

where $\mathbf{e}(a_t^*)$ is the one-hot encoding of the accepted action. The loss is applied only when an accepted candidate exists. We distill toward the accepted action with an MSE-to-one-hot target rather than cross-entropy: regressing the logits to $\mathbf{e}(a_t^*)$ gives a bounded, low-variance teaching signal that we found more stable under the sparse accepted-candidate supervision.

Forecaster anchoring and bootstrap regularization. During Stage II, the forecaster continues to receive factual supervision through $\mathcal{L}_{\text{fr}}^{\text{fact}}$. For accepted candidates, we further add a confidence-gated bootstrap regularization term:

$$\mathcal{L}_{\text{boot}} = \left\| F_\phi^s(s_t, a_t^*) - \mathrm{sg}\left[\bar{F}_\phi^s(s_t, a_t^*)\right] \right\|_2^2, \tag{69}$$

where $\bar{F}_\phi^s$ is the MC-dropout averaged state prediction obtained during candidate evaluation, and $\mathrm{sg}[\cdot]$ stops gradients. Intuitively, $\mathcal{L}_{\text{boot}}$ is a confidence-gated variance-reduction anchor: since the counterfactual action has no factual next-state label, it pulls a single stochastic forward pass toward its own MC-dropout consensus $\bar{F}_\phi^s$, stabilizing the forecaster on counterfactual inputs only when uncertainty is low. The confidence score and bootstrap weight are defined as

$$c_t = \mathrm{clip}\left(1 - \frac{\hat{\sigma}_\phi(s_t, a_t^*)}{\sigma_{\text{thr}}}, 0, 1\right), \tag{70}$$

$$\alpha_t = \mathbb{I}[c_t \geq \kappa] \cdot c_t \cdot \max\left(\alpha_{\min}, 1 - \frac{n}{N}\right), \tag{71}$$

where $\kappa = 0.5$, $\alpha_{\min} = 0.3$, $N = 5\times10^4$, and n is the global training step.

The Stage II objective is

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{cf}} + 0.5\left(\mathcal{L}_{\text{fr}}^{\text{fact}} + \alpha_t \mathcal{L}_{\text{boot}}\right). \tag{72}$$

Thus, the policy is updated only through accepted counterfactual candidates, while the forecaster remains anchored to factual transitions and receives bootstrap regularization only under sufficient confidence.

7.5 Inference Closed-Loop Semantic-Guided Treatment Rollout

At inference time, ClinSDT generates treatment actions in a closed-loop rollout. The current state is first converted into Profile, Context, and Advisory descriptions according to the information available at that step. These descriptions are encoded by the frozen LLM and passed to the policy model. The current action token is absent, and the policy selects the treatment action by

$$\hat{a}_t = \arg\max_{a\in\mathcal{A}} \ell_t[a], \tag{73}$$

from the input

$$[z_t^p, R_t, z_t^c, z_t^a, s_t]. \tag{74}$$

For rollout beyond the initially observed state, future acuity annotations are not available. Therefore, ClinSDT uses a state-only semantic template for predicted states, constructed from the physiological state variables rather than future labels. Given the current rollout state $\hat{s}_k$ and policy action $\hat{a}_k$, the forecaster updates the state as

$$\hat{s}_{k+1} = F_\phi^s(\hat{s}_k, \hat{a}_k). \tag{75}$$

The updated state is then used to refresh the semantic inputs for the next decision step. This procedure is repeated for the rollout horizon. Throughout inference, the LLM only encodes semantic descriptions, while treatment actions are generated by the policy model.

8 Experiments

We evaluate ClinSDT from four perspectives that follow the motivation of the paper: whether structured clinical semantics improve offline policy learning, which semantic designs contribute to the gain, whether semantic tokens participate in policy learning and counterfactual refinement, and how the rollout-based evidence should be interpreted.

RQ1: Does structured LLM-encoded clinical semantics improve offline sepsis treatment policies?

RQ2: Which semantic factors matter: the clinical views themselves or their token organization?

RQ3: Does Stage II provide additional counterfactual refinement, and are the semantic tokens actively used by the policy?

RQ4: What are the reliability boundaries of model-based rollout evaluation?

Datasets. We evaluate on offline ICU sepsis treatment trajectories from MIMIC-III and MIMIC-IV [6, 7]. Each trajectory contains physiological measurements, historical treatment actions, and subsequent patient-condition changes. Following common sepsis treatment settings, each state is represented by a 45-dimensional physiological feature vector, and actions are discretized according to intravenous fluid and vasopressor levels. For subgroup analysis, patients are divided into **HIGH**, **MID**, and **LOW** acuity groups according to initial SAPS-II scores. Because SAPS-II distributions differ across datasets, we use dataset-specific thresholds while keeping the same three-level acuity structure.

Evaluation metric. We use SAPS-II change as the proxy outcome for model-based rollout evaluation. Let $q_{i,t}$ denote the predicted SAPS-II score of trajectory i at rollout step t. For horizon K, the cumulative SAPS-II change is

$$\Delta\text{SAPS}_{i,K} = \sum_{t=1}^{K}(q_{i,t} - q_{i,t-1}) = q_{i,K} - q_{i,0}. \tag{76}$$

Since lower SAPS-II indicates improved patient condition, we report SAPS-II improvement:

$$\text{Imp}_{i,K} = -\Delta\text{SAPS}_{i,K}. \tag{77}$$

The final score is averaged over the test set $\mathcal{D}$:

$$\text{Imp}_K = \frac{1}{|\mathcal{D}|}\sum_{i\in\mathcal{D}} \text{Imp}_{i,K}. \tag{78}$$

Higher values indicate better predicted outcomes. We report both 5-step and 10-step rollouts. All policies are evaluated with the same learned rollout environment and the same initial test trajectories. Therefore, this metric should be interpreted as controlled offline evidence for relative policy comparison, rather than as direct evidence of clinical efficacy.

Baselines. We compare ClinSDT with representative baselines from imitation learning, value-based RL, offline RL, and sequence-modeling-based policy learning. **BC** [13] imitates logged clinician actions. **DQN** [12] is a value-based RL baseline trained under the same offline data and rollout protocol. **BCQ** [3] constrains action selection to the support of the behavior data. **IQL** [10] performs implicit offline policy optimization without explicit out-of-distribution action evaluation. **DT** [2] is a return-conditioned Decision Transformer trained on return, state, and action sequences.

8.1 Overall Effectiveness of Structured Clinical Semantics (RQ1)

RQ1 evaluates whether adding structured clinical semantics to a DT-style policy improves offline sepsis treatment decision-making. Table 2 reports SAPS-II improvement on MIMIC-III and MIMIC-IV under 5-step and 10-step rollouts.

ClinSDT achieves the best ALL-patient performance on both datasets and both rollout horizons. On MIMIC-III, the strongest baseline reaches +0.57 under the 5-step rollout and +2.94 under the 10-step rollout, while ClinSDT achieves +1.36 and +8.75, respectively. The gain is more visible in the longer rollout, where policy decisions interact with evolving patient states over multiple steps. Under the 10-step setting on MIMIC-III, ClinSDT also

Table 2: Main rollout results on MIMIC-III and MIMIC-IV. We compare ClinSDT with representative baselines from imitation learning, offline RL, value-based RL, and Decision Transformer-style policy learning. The metric is SAPS-II improvement, defined as the negative cumulative SAPS-II change over rollout trajectories. Higher values indicate better predicted outcomes. Bold indicates the best result in each column, and underlining indicates the second-best result.

Dataset	Model	5-step rollout				10-step rollout			
		HIGH	MID	LOW	ALL	HIGH	MID	LOW	ALL
MIMIC-III	BC	3.24	-1.51	-5.59	-0.59	5.39	-1.55	-7.80	-0.22
	BCQ	3.17	-1.30	-3.88	-0.35	5.99	-1.17	-5.59	0.34
	DQN	6.42	-1.05	-5.03	0.57	13.17	0.10	-6.74	2.94
	DT	**7.01**	-1.57	-5.93	0.30	12.96	-1.40	-8.98	1.72
	IQL	2.86	-2.13	-6.13	-1.14	4.39	-2.58	-9.25	-1.28
	Ours	6.91	**-0.44**	**-1.43**	**1.36**	**17.74**	**5.16**	**10.77**	**8.75**
MIMIC-IV	BC	0.55	-0.80	-0.62	-0.26	0.34	-1.61	-1.23	-0.79
	BCQ	1.82	-0.66	-0.93	0.13	**2.59**	-1.02	-1.53	0.10
	DQN	1.38	-0.43	-0.50	0.19	1.97	-0.95	-1.06	0.06
	DT	**2.17**	-0.74	-0.95	0.23	2.48	-1.90	-2.18	-0.43
	IQL	0.17	-0.84	-0.77	-0.46	-1.23	-2.15	-1.62	-1.65
	Ours	0.65	**0.21**	**0.17**	**0.35**	0.76	**0.30**	**0.27**	**0.45**

achieves the best result in all three acuity groups, suggesting that the overall gain is not driven by a single subgroup.

On MIMIC-IV, the absolute improvements are smaller, but ClinSDT still achieves the best ALL score under both horizons, reaching +0.35 and +0.45 for 5-step and 10-step rollouts. This cross-dataset trend supports the main hypothesis of the paper: raw physiological trajectories alone may leave important decision semantics implicit, while structured clinical semantic tokens provide useful context for sequential policy learning.

These results should be interpreted as relative policy-quality evidence under a shared learned evaluator. They do not establish real-world clinical effectiveness, but they indicate that the proposed semantic-guided policy consistently compares favorably with imitation learning, offline RL, and standard DT baselines in the controlled rollout setting. Figure 6 further summarizes the ALL-patient results across datasets and rollout horizons.

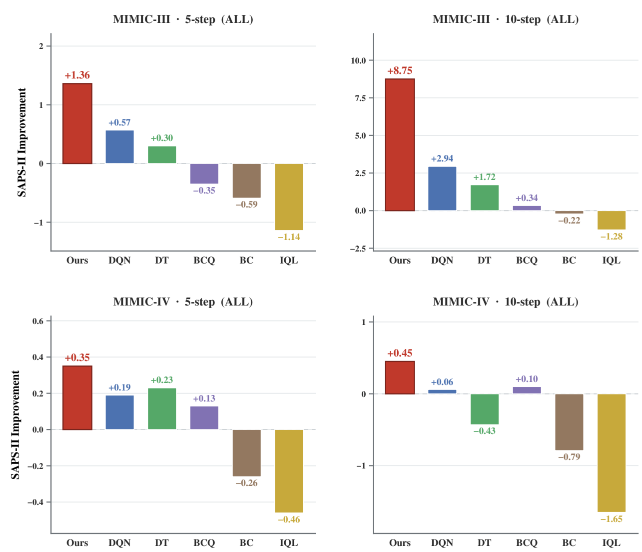

Figure 6: Main rollout results on MIMIC-III and MIMIC-IV. ClinSDT achieves the best ALL-patient performance under both 5-step and 10-step rollouts, with the clearest gain on the longer MIMIC-III rollout.

8.2 Semantic Factors and Token Organization (RQ2)

RQ2 examines which part of the semantic design contributes to the observed gains. We separate two factors: the content of the clinical semantic views and the way these views are organized in the policy input. This distinction is important because ClinSDT does not simply concatenate extra features; it inserts role-specific semantic views as separate tokens.

Semantic-view variants. Table 3 reports removal and single-view variants on MIMIC-III. The full model obtains the best 10-step ALL score of +8.75. Removing Profile, Context, or Advisory reduces the score to +0.84, +1.37, and +1.10, respectively. This suggests that the three views support the main metric in different ways: Profile anchors overall condition, Context captures recent response, and Advisory highlights risk priorities.

Single-view variants show a more mixed pattern. Using only one view can help in some subgroups or horizons, but is less stable than the full design. For example, only Context performs strongly on the 10-step HIGH group, whereas the full model gives the best 10-step ALL result and the strongest MID/LOW performance. Thus, the full design is not uniformly best in every subgroup, but is more stable on the main longer-horizon metric.

Table 3: Semantic-view variants on MIMIC-III under 5-step and 10-step rollouts. Results are SAPS-II improvements, where higher values indicate better outcomes. Best and second-best results are highlighted in bold and underlined.

Group	Variant	5-step rollout				10-step rollout			
		HIGH	MID	LOW	ALL	HIGH	MID	LOW	ALL
Removal	w/o Profile	3.14	-0.54	-4.96	0.09	2.52	0.30	-0.03	0.84
	w/o Context	5.66	-0.43	-4.97	0.80	5.92	-0.14	-0.68	1.37
	w/o Advisory	5.97	-1.30	-7.94	0.09	4.26	0.05	-0.28	1.10
	w/o Semantics	7.06	-2.48	-9.69	-0.56	4.82	0.41	0.39	1.53
Single-view	only Profile	<u>8.45</u>	-0.79	-3.67	<u>1.36</u>	1.68	-0.94	-0.66	-0.25
	only Context	2.52	<u>-0.13</u>	-6.42	0.10	**24.12**	<u>1.56</u>	<u>1.75</u>	<u>7.31</u>
	only Advisory	**9.19**	**0.71**	**-0.43**	**2.79**	6.28	0.43	0.43	1.91
Full	**Ours**	6.91	-0.44	<u>-1.43</u>	<u>1.36</u>	<u>17.74</u>	**5.16**	**10.77**	**8.75**

Table 4: Comparison of token-fusion designs on MIMIC-III under the 10-step rollout. Higher SAPS-II improvement indicates better outcomes.

Variant	Design	HIGH	MID	LOW	ALL
Mean	zero-param pooling	0.49	-0.24	-1.78	-0.16
Concat	one projection	0.62	-0.35	-7.43	-0.60
Gated	two projections	4.08	-1.19	-21.50	-1.28
Residual	residual projection	-0.02	-1.34	-6.78	-1.38
Cross-attn	MHA + LN	-0.94	-1.48	-5.89	-1.65
Ours	6 separated tokens	**17.74**	**5.16**	**10.77**	**8.75**

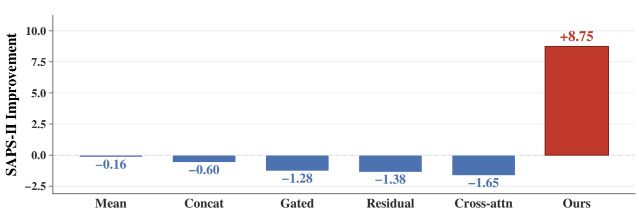

Figure 7: Token-fusion variants on MIMIC-III. Keeping semantic views as separate tokens yields the strongest 10-step SAPS-II improvement, suggesting that role-specific token organization matters.

Token-fusion variants. Table 4 compares different ways of integrating the semantic views. The proposed six-token design substantially outperforms mean pooling, concatenation, gated fusion, residual fusion, and cross-attention fusion. These alternatives compress or mix the semantic roles before sequence modeling, while ClinSDT keeps Profile, Context, and Advisory as independent tokens. The result suggests that the gain is not only due to adding LLM-derived embeddings, but also depends on preserving their role-specific structure within the DT input sequence. Figure 7 visualizes the same comparison and highlights the advantage of keeping semantic views as separate tokens.

Figure 8: Stage I to Stage II rollout change on MIMIC-III. The y-axis reports ΣΔSAPS-II, where lower values indicate better predicted outcomes.

Table 5: Stage II improvement over the corresponding Stage I policy on MIMIC-III.

Model	5-step improvement				10-step improvement			
	HIGH	MID	LOW	ALL	HIGH	MID	LOW	ALL
DQN	2.93	0.45	0.28	1.06	6.76	1.41	1.07	2.75
DT	3.79	0.76	1.28	1.56	7.03	1.29	1.86	2.79
BCQ	1.21	1.38	1.71	1.36	2.70	2.15	2.58	2.32
BC	0.08	0.49	0.46	0.39	0.79	0.86	0.62	0.83
IQL	0.11	0.43	0.24	0.33	0.74	0.87	0.68	0.83
Ours	2.03	−0.06	0.47	0.51	3.26	3.63	2.94	3.49

8.3 Counterfactual Refinement and Semantic-Token Diagnostics (RQ3)

RQ3 contains two complementary analyses. First, we evaluate whether Stage II counterfactual semantic distillation provides additional refinement beyond the Stage I policy. Second, we use ablations and attention patterns to diagnose whether the semantic tokens participate in the learned policy behavior.

Counterfactual refinement. Figure 8 visualizes the change from Stage I to Stage II on MIMIC-III. After counterfactual semantic distillation, ClinSDT further reduces the predicted cumulative SAPS-II change and maintains the lowest final value among the compared policies. This indicates that Stage II provides additional forecaster-guided refinement beyond the Stage I semantic-token policy.

Table 5 further quantifies the Stage II improvement over the corresponding Stage I policy. ClinSDT achieves the largest 10-step ALL improvement of +3.49, with clear gains in the MID and LOW groups. The 5-step results are more mixed, where DT obtains the highest ALL improvement. This pattern is consistent with counterfactual refinement being more visible when predicted effects accumulate over a longer horizon, while short rollouts are more affected by local imitation.

Semantic-token diagnostics. We next examine whether the semantic tokens participate in the learned policy behavior. The first evidence comes from the semantic-view variants in Table 3: removing individual semantic views weakens the full model under the main 10-step ALL metric. This suggests that the semantic views are not simply redundant appended inputs.

We further inspect attention allocation in the Stage II policy. As shown in Table 6, the pattern varies across acuity groups. For HIGH-acuity patients, Profile, Advisory, and the aggregated semantic group receive positive relative deviations, suggesting greater attention to severity and risk-priority information. For LOW-acuity patients, attention shifts relatively more toward RTG and State tokens.

We do not interpret attention as causal proof. Rather, it provides a diagnostic view that complements the semantic-view variants.

Table 6: Stratum-wise relative attention scores of the Stage II policy on MIMIC-III. Positive values indicate higher relative attention to the corresponding token group, while negative values indicate lower relative attention.

Stratum	Profile	Context	Advisory	Sem.	RTG	State
HIGH	+0.0031	−0.0016	+0.0014	+0.0029	+0.0013	−0.0043
MID	−0.0012	+0.0007	−0.0005	−0.0010	−0.0007	+0.0016
LOW	−0.0021	−0.0003	−0.0018	−0.0042	+0.0016	+0.0025

Figure 9: Stratum-wise relative attention heatmap of the Stage II policy on MIMIC-III. Warmer and cooler colors indicate higher and lower relative attention, respectively.

Together, these results suggest severity-dependent use of the semantic structure, while the main evidence for semantic contribution comes from the ablation results. Figure 9 provides a heatmap visualization of the same stratum-wise attention pattern.

8.4 Reliability Boundaries of Model-based Rollout (RQ4)

RQ4 examines how the model-based rollout results should be interpreted. Since online interaction with ICU patients is infeasible, we use a learned Treatment-Response Forecaster for controlled policy comparison [18]. Thus, the rollout results provide offline relative evidence rather than clinical validation.

We assess this boundary from two aspects. First, we compare rollout horizons. The main evaluation uses $K = 10$ to capture multi-step treatment effects, while $K = 5$ serves as a shorter-horizon check. This test examines whether the relative trend remains stable when the rollout horizon changes, rather than identifying a clinically optimal horizon.

Second, we evaluate whether the Stage II policy remains within behavior support. As shown in Table 7, the overall OOD ratio is low (4.84%), and the predicted action dispersion closely matches the behavior dispersion ($\sigma_{\text{pred}} = 0.937$ vs. $\sigma_{\text{beh}} = 0.938$). This suggests that the rollout improvement is not mainly driven by broad deviation from the offline data support.

The LOW stratum shows weaker support matching, with higher KL divergence (0.883) and a slightly higher OOD ratio (5.17%). Therefore, LOW-stratum results should be interpreted more cautiously. Overall, ClinSDT shows stronger relative rollout performance while largely preserving behavior-support consistency, with conclusions bounded by the learned forecaster and offline data coverage.

Table 7: Detailed behavior-support statistics of the Stage II policy on MIMIC-III.

Group	n	Top1	σ_{beh}	σ_{pred}	OOD%	KL
High	8362	0.458	0.910	0.907	4.87	0.662
Mid	18646	0.483	0.943	0.942	4.93	0.712
Low	1914	0.399	1.010	1.012	5.17	0.883
Overall	28922	0.470	0.938	0.937	4.84	0.685

9 Conclusion

We presented **ClinSDT**, an LLM-assisted Decision Transformer framework for offline sepsis treatment. ClinSDT uses a frozen LLM to encode PROFILE, CONTEXT, and ADVISORY semantics as independent policy tokens, while treatment actions are generated only by the learned policy. With Treatment-Response Forecasting and conservative counterfactual distillation, ClinSDT introduces structured clinical context into offline policy learning.

Experiments on MIMIC-III and MIMIC-IV show consistent relative gains over representative baselines under the same rollout protocol. Further analyses support the contribution of semantic views, token-level organization, and confidence-filtered counterfactual refinement. These results suggest that LLM-encoded clinical semantics can help expose implicit decision context, while the conclusions remain bounded by model-based rollout evaluation.

GenAI Usage Disclosure

Generative AI tools were used only to assist with language polishing and clarity improvement during manuscript preparation. The authors are responsible for all technical content, experimental design, implementation, analysis, and conclusions. No generative AI tool was used to produce experimental results, fabricate data, generate references, or make clinical treatment decisions.

The use of a frozen large language model in ClinSDT is part of the proposed method. It is used solely as a semantic encoder for rule-constructed clinical descriptions, while treatment actions are generated by the policy model.

A Dataset, Implementation, and Evaluation Details

This appendix summarizes the dataset construction, semantic templates, model configuration, and evaluation protocol used in our experiments.

We use dataset-specific SAPS-II thresholds for stratified analysis because the initial SAPS-II distributions differ substantially between MIMIC-III and MIMIC-IV.

B Clinical Semantic Templates

ClinSDT uses the frozen LLM only as a semantic encoder. The semantic descriptions are generated by rule-based templates and are inserted into the policy as independent tokens.

Table 8: Dataset construction and preprocessing details.

Item	Setting
Datasets	MIMIC-III and MIMIC-IV
Cohort construction	AI Clinician sepsis pipeline
Inclusion criterion	Adult ICU stays satisfying Sepsis-3 criteria
Time discretization	4-hour intervals
State space	45-dimensional physiological state
State variables	Demographics, vitals, labs, fluid I/O, ventilation status
Missing values	LOCF and population-median imputation
Normalization	Z-score normalization using training-set statistics
Action space	5×5 fluid–vasopressor bins, 25 discrete actions
Action encoding	$a = 5b_{\text{fluid}} + b_{\text{vaso}}$
Reward	$r_t = -(\text{SAPS-II}_{t+1} - \text{SAPS-II}_t)$
RTG discount / scale	$\gamma = 0.99$ / 10

Table 9: Dataset statistics and rollout eligibility. A trajectory is eligible for the 10-step rollout if it contains at least 11 time steps.

Item	MIMIC-III	MIMIC-IV
Train trajectories	13,011	52,380
Validation trajectories	2,781	11,225
Test trajectories	2,786	11,224
10-step eligible test trajectories	1,966	8,153
State dimension	45	45
Action dimension	25	25
Context length	20	20
Rollout horizon	10	10
RTG discount / scale	0.99 / 10	0.99 / 10
Trajectory length	2–20, mean 13.4	6–768, mean 26.9
Semantic embedding dimension	896 per view	896 per view

Table 10: Initial SAPS-II distribution and acuity stratification. Counts are computed on the 10-step eligible test subset.

Item	MIMIC-III	MIMIC-IV
Initial SAPS-II mean	81.8	53.5
Initial SAPS-II range	65.5–91.5	37–86
Initial SAPS-II q10/q50/q90	79.5 / 80.5 / 91.5	44 / 53 / 64
Low threshold	< 75	< 50
Mid threshold	$[75, 85)$	$[50, 57)$
High threshold	≥ 85	≥ 57
High count	500	2,909
Mid count	1,328	2,544
Low count	138	2,700

Table 11: Clinical semantic views used by ClinSDT.

View	Input	Role
Profile	s_t	Current severity, organ burden, ventilation, and target
Context	(s_{t-1}, a_{t-1}, s_t)	Recent treatment-response transition
Advisory	s_t	Current risk priorities without action recommendation

C Model and Hyperparameters

For semantic-fusion ablations on MIMIC-III, we evaluate Concat, Residual, Gated, and Cross-Attention variants. These variants use three semantic tokens per step, resulting in a block size of $20 \times 3 = 60$.

Table 12: Template regimes used to avoid future-label leakage.

Regime	Used in	Information source
Acuity-enriched templates	Factual training	Observed states with recorded acuity information
State-only templates	Counterfactual branches and rollout	45-dimensional physiological state only

Table 13: Example semantic descriptions.

View	Example
Profile	Severe sepsis patient with significant organ involvement. Severity: high. Multi-organ involvement: cardiovascular and renal. Patient is on mechanical ventilation. Target: cumulative SAPS-II improvement of 6.0 points.
Context	After vasopressor titration up: blood pressure is stable. Lactate is falling, perfusion improving. Renal function is stable.
Advisory	Lactate is critically high, indicating severe tissue hypoperfusion. Overall risk is high with significant organ dysfunction. Focus on cardiovascular protection.

Table 14: Model architecture and main hyperparameters. The same settings are used for MIMIC-III and MIMIC-IV unless otherwise stated.

Item	Value
Policy backbone	Causal Transformer
Token sequence per step	[Profile, RTG, Context, Advisory, s_t, a_t]
Action prediction position	State-token hidden state
Policy layers / heads / hidden size	4 / 8 / 128
Context length K	20
Tokens per step	6
Block size	$20 \times 6 = 120$
LLM embedding dimension	896 per view
Forecaster input	Numerical state–action pair (s_t, a_t)
Forecaster output	$(\mu_{t+1}, \log \sigma_{t+1}, \widehat{\Delta\text{SAPS}_t})$
Forecaster type / hidden size	3-layer MLP / 256
MC Dropout rate / samples	0.2 / 10
Policy learning rate	6×10^{-4}
Forecaster learning rate	1×10^{-3}
Batch size	64
Weight decay	0.1
Stage I / Stage II epochs	100 / 50
Stage II iterations	1000
Search radius	2
Uncertainty threshold σ_{thr}	2.0
Confidence gate κ	0.5
Bootstrap floor $\alpha_{\min}$	0.3
Forecaster loss weight λ_o	0.5

Table 15: Training and evaluation protocol.

Stage	Protocol
Stage I policy learning	Train the semantic-token Decision Transformer by behavior cloning.
Stage I forecaster learning	Train the Treatment-Response Forecaster on factual state–action transitions.
Stage II candidate generation	Search local counterfactual actions around the documented action and include the current policy action.
Stage II screening	Accept candidates only if uncertainty is below σ_{thr} and the predicted SAPS-II increase is lower than that of the documented action.
Stage II distillation	Regenerate Context and Advisory from the predicted counterfactual transition and distill accepted candidates into the policy.
Evaluation rollout	Start from a real test state, predict the policy action, update the next state with the shared forecaster, and append the predicted state to the decision context.
Reported metric	SAPS-II improvement, $-\sum_t \Delta\text{SAPS}_t$; higher is better.
Interpretation	Controlled relative comparison under a shared learned rollout model, not direct clinical efficacy evidence.

References

[1] Emily Alsentzer, John Murphy, William Boag, Wei-Hung Weng, Di Jindi, Tristan Naumann, and Matthew McDermott. 2019. Publicly available clinical BERT embeddings. In *Proceedings of the 2nd clinical natural language processing workshop*. 72–78.

[2] Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Misha Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. 2021. Decision transformer: Reinforcement learning via sequence modeling. *Advances in neural information processing systems* 34 (2021), 15084–15097.

[3] Scott Fujimoto, David Meger, and Doina Precup. 2019. Off-policy deep reinforcement learning without exploration. In *International conference on machine learning*. PMLR, 2052–2062.

[4] Richard S Hotchkiss and Irene E Karl. 2003. The pathophysiology and treatment of sepsis. *New England journal of medicine* 348, 2 (2003), 138–150.

[5] Michael Janner, Qiyang Li, and Sergey Levine. 2021. Offline reinforcement learning as one big sequence modeling problem. *Advances in neural information processing systems* 34 (2021), 1273–1286.

[6] Alistair EW Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, et al. 2023. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific data* 10, 1 (2023), 1.

[7] Alistair EW Johnson, Tom J Pollard, Lu Shen, Li-wei H Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G Mark. 2016. MIMIC-III, a freely accessible critical care database. *Scientific data* 3, 1 (2016), 1–9.

[8] Matthieu Komorowski, Leo A Celi, Omar Badawi, Anthony C Gordon, and A Aldo Faisal. 2018. The artificial intelligence clinician learns optimal treatment strategies for sepsis in intensive care. *Nature medicine* 24, 11 (2018), 1716–1720.

[9] Guilan Kong, Dong-Ling Xu, and Jian-Bo Yang. 2008. Clinical decision support systems: a review on knowledge representation and inference under uncertainties. *International Journal of Computational Intelligence Systems* 1, 2 (2008), 159–167.

[10] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. 2021. Offline reinforcement learning with implicit q-learning. *arXiv preprint arXiv:2110.06169* (2021).

[11] Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. 2020. Offline reinforcement learning: Tutorial, review, and perspectives on open problems. *arXiv preprint arXiv:2005.01643* (2020).

[12] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. 2015. Human-level control through deep reinforcement learning. *nature* 518, 7540 (2015), 529–533.

[13] Dean A Pomerleau. 1988. Alvinn: An autonomous land vehicle in a neural network. *Advances in neural information processing systems* 1 (1988).

[14] Aravind Rajeswaran, Vikash Kumar, Abhishek Gupta, Giulia Vezzani, John Schulman, Emanuel Todorov, and Sergey Levine. 2017. Learning complex dexterous manipulation with deep reinforcement learning and demonstrations. *arXiv preprint arXiv:1709.10087* (2017).

[15] Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, et al. 2023. Large language models encode clinical knowledge. *Nature* 620, 7972 (2023), 172–180.

[16] Karan Singhal, Tao Tu, Juraj Gottweis, Rory Sayres, Ellery Wulczyn, Mohamed Amin, Le Hou, Kevin Clark, Stephen R Pfohl, Heather Cole-Lewis, et al. 2025. Toward expert-level medical question answering with large language models. *Nature medicine* 31, 3 (2025), 943–950.

[17] Shengpu Tang and Jenna Wiens. 2021. Model selection for offline reinforcement learning: Practical considerations for healthcare settings. In *Machine Learning for Healthcare Conference*. PMLR, 2–35.

[18] Philip Thomas, Georgios Theocharous, and Mohammad Ghavamzadeh. 2015. High confidence policy improvement. In *International Conference on Machine Learning*. PMLR, 2380–2388.

[19] Da Wang, Lin Li, Wei Wei, Qixian Yu, Jianye Hao, and Jiye Liang. 2025. Improving generalization in offline reinforcement learning via latent distribution representation learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 21053–21061.

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